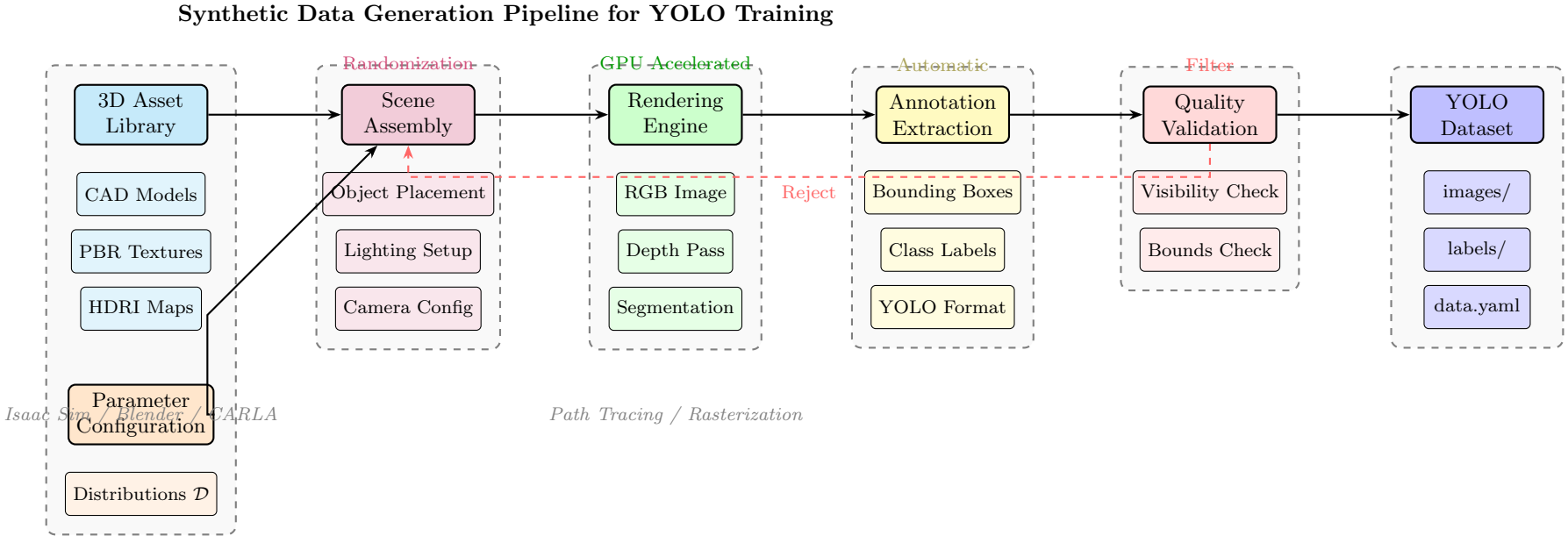
Synthetic Data Generation Pipeline for YOLO Training
3D Asset Library
CAD Models
PBR Textures
HDRI Maps
Parameter Configuration
Distributions $\mathcal{D}$
Isaac Sim / Blender / CARLA
Randomization
Scene Assembly
Object Placement
Lighting Setup
Camera Config
GPU-Accelerated
Rendering Engine
RGB Image
Depth Pass
Segmentation
Path Tracing / Rasterization
Automatic
Annotation Extraction
Bounding Boxes
Class Labels
YOLO Format
Filter
Quality Validation
Visibility Check
Bounds Check
Reject
YOLO Dataset
images/
labels/
data.yaml